

Effect of *Trichoderma koningii* on the Growth Yield of *Capsicum chinense* Jacq. (NHCC-AC9) Against *Fusarium oxysporum* and *Pythium ultimum*

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Abstract

Biological control of pests and diseases of *Capsicum* plants and crops of economic importance has been a major solution to the challenges faced by farmers and the entire population of the world. *Capsicum chinense* is known to be affected by pathogenic organisms that reduce the quality and quantity of its production. *Trichoderma koningii* (bioagent) was used as a biocontrol agent to reduce the infestations of the soil borne disease of *Capsicum*. The varieties used for this research were obtained from National Horticultural Research Institute (NIHORT), Ibadan. The bioagent (T) and pathogens (P1 and P2) were prepared into two concentrations, C1 (1.0×10^5 spore/mL) and C2 (1.0×10^7 spore/mL) respectively. Exactly 20mL each was inoculated into the soil at 24hours intervals before planting. Measurement of growth parameters was taken once a week after 2 weeks of planting for 10 weeks and diseased severity scale (1-7) was observed. This research work showed that there was a significant ($P < 0.05$) increase in the following growth parameters: plant heights, leaf area, stem girth, number of leaves, petiole length, branch number, and nodal length. The diseased severity scale value of zero (0) was observed in *Capsicum chinense* treated with TC1 and TC2, while the plant treated with pathogens at higher concentrations P1C2 and P2C2 had the highest diseased severity scale of 4 and 5 respectively. The use of *T.koningii* as alternative means of combating *Capsicum* diseases have been proved to be effective, economic and environmental friendly.

1.0 Introduction

The genus *Capsicum* is part of the large solanaceae family, which, among the more than 90 genera and 2500 species of flowering plants, includes commercially important vegetables such as tomato, potato, and eggplant. [1]. At the beginning, fruits were exchanged for black pepper [2]. *Piper nigrum* is a species similar in taste though not in appearance and not phylogenetically related to *Capsicum*. For this reason, it was incorrectly named “pepper” [3][4].

Soilborne fungal diseases are major pathogens in *Capsicum* production in Nigeria.

Pythium spp. *Fusarium* spp., *Alternaria* spp., and *Phytophthora* spp. are mostly associated with vegetable crops especially pepper. These pathogens can cause more than 60% seedling mortality in nursery, screenhouse and in the field [5].

Pre-emergence and post-emergence damping-off diseases occur in plants susceptible to *Pythium* spp. *Pythium ultimum* is one of the most destructive among *Pythium* species that causes seed and seedling losses [6]. *Pythium* species responsible for pre-emergence damping-off attack seeds or seedlings before emergence, while post-emergence damping-off causes decay and death of seedlings after emergence [7]. *Pythium* spp. can also cause wilt in older seedling and mature plants as well.

Fusarium species are known to also cause significant reduction in yield of crops. *F. oxysporum*, *F. proliferatum*, *F. solani*, *F. moliniforme*, *F. anthophilum*, *Macrophomina phaseolina*, and *Rhizoctonia solani* were known worldwide to be associated with wilt disease symptoms [8], but the most common causal agent of wilt disease are *F. oxysporum* and *F. solani*. Among *Fusarium* spp. wilting it is mostly associated with *F. oxysporum* [8].

Traditionally, *Capsicum* farmers relied on chemical pesticides and fertilizers for optimum yields, with undesirable effects to the ecosystem and human health. Despite an increase in fertilizer and pesticide usage, there is still a decline in crop yield as a result of pests and diseases.

Biological control offers a potential alternative control measure in the area of plant diseases and becomes a promising tool to use in agricultural production as it reduces the release of polluting chemical pesticides to the ecosystem [9]. At present, quite a number of biocontrol agents (BCA) are found to be able to manage the plant diseases effectively as well as being environmental friendly [10]. Many biopesticide and biofertilizer products are now available in the market, the majority of which are based on beneficial symbionts of the genus *Trichoderma*[11]. These fungi are well known for

their ability to kill plant pathogens [1], [12], promote plant growth and resistance against biotic and abiotic stresses. *Trichoderma* spp. have evolved numerous mechanisms that are involved in attacking other fungi. These mechanisms include competition for space and nutrients, mycoparasitism[13], production of inhibitory compounds, inactivation of the pathogen's enzymes and induced plant resistance [14]. Therefore, the aim of this research is to evaluate the effects of *Trichoderma koningii* against wilting and root rot pathogens. The physiological parameters, disease severity, yield components and phytochemical components of the leaves of *Capsicum chinense* were recorded to evaluate their performance in the presence of *Fusarium oxysporum* and *Pythium ultimum* under green house.

Materials and Methods

2.1 Study site

This experiment was carried out in a greenhouse located at the Federal University Oye-Ekiti, Department of Plant Science and Biotechnology, with Latitude 7°, 28'N and Longitude 5°, 44'E. The temperature of the screen house ranged between 25°C-35°C. The experiment was conducted within the months of May-August 2021.

2.2 Collection of Seeds

The variety of *Capsicum* used in the experiment was collected from National Horticultural Research Institute (NIHORT), Ibadan, Oyo state.

2.3 Stock culture of *Trichoderma*

The pure culture of *Trichoderma koningii* used in the research work was collected from the plant pathology/mycology units of the Department of Plant Science and Biotechnology, Federal University Oye-Ekiti, Ekiti State.

2.4 Collection of soil samples

A sandy loamy soil sample collected around the university premises, and sterilized in an oven pre-set at 103°C were used for the experiment. The sterile soil was potted into a perforated potting bags and moved to the screenhouse immediately.

2.5 Antagonistic effect of *Trichoderma koningii* against *Fusarium oxysporum* and *Pythium ultimum* as observed in *Capsicum chinense* in-vivo.

The antagonistic capabilities of *Trichoderma koningii* against *Fusarium oxysporum* was studied in the greenhouse of the Department of Plant Science and Biotechnology, Federal University Oye-Ekiti, Ekiti State, Nigeria. The potted soil was labelled and arranged according to the treatment combinations spaced at 25cm apart. The thermometer was used to monitor the temperature in the greenhouse. Exactly 10 mL of the pathogen was inoculated into the soil, after which the inoculation of the antagonist was applied into the soil at 24 hours interval. The plants were watered every other day and observed for growth daily. Measurement of growth parameters were taken after 2 weeks of planting.

2.6 Physiological parameters

Growth parameters such as plant height, shoot length, branch number, leaf number, leaf area, petiole length, stem girth, and nodal length of the *Capsicum* plants were monitored. The measurement were first taken after two weeks of planting, and subsequently taken at interval of one week for the period of eight weeks.

2.6.1 Plant height

The plant height was measured and determined by using meter rule to measure the height of the *Capsicum* plant from the soil level to the terminal bud.

2.6.2 Number of leaves

The leaf numbers were determined by physical counting of the leaves on a *Capsicum* plant and the observed values recorded.

2.6.3 Leaf area

The length and breadth of the randomly selected leaf were measured using a linear meter rule, the mean values of the length and the breadth were multiplied together to get the leaf area in cm²

2.6.4 Petiole length

The petiole length was measured by also using a meter rule to measure the distance between the leaf base and the point of attachment to the stem.

2.6.5 Stem girth

The stem girth was measured with the use of a manual vernier calliper at the basal region of the stem, it was done 2cm above the soil level.

2.6.6 Number of branches

The number of branches on each plant were counted manually.

2.6.7 Internode length

The distance between two nodes on the stem of *Capsicum chinense* plant was measured using meter rule

2.7 Preparation of plant extract

The leaves were detached from selected *Capsicum* plants and washed under running tap water to remove dirt. The leaf samples were then air-dried for few days and the leaves were grinded into powder form. Hundred grams of powdered material was soaked in 200mL ethanol in McCartney bottles with thorough shaking in a shaker incubator for 24hrs at 28°C. The extract was then filtered through a filter paper and the concentrated extract was dried at room temperature. The extract was used for determination of phytochemical analysis.

2.8 Phytochemical Analysis

The phytochemical analysis of *Capsicum chinense* leaves was carried out according to standard methods [9].

2.8.1 Test for Saponins

Exactly 20mL of distilled water was added to 5mL of solvent extract in a test-tube with intermittent shaking in a 100mL cylinder for 20 minutes. Formation of a layer of foam indicate the presence of saponins [1].

2.8.2 Test for Flavonoids

Flavonoid test was done by adding few drops of concentrated ammonia to a portion of the extract in a test-tube. A yellow colouration indicates the presence of flavonoids [15].

2.8.3 Test for Alkaloids

About 5 mL of 2% HCl was added to 5mL of the extract, then 3mL of Mayer's reagent was also added. The mixture was heated in the test tube, filtered and then few drops of picric acid was poured to the filtrate. The formation of yellow colour precipitate indicates the presence of alkaloids [16].

2.8.4 Test for Terpenoids

To 2 mL of the extract in a test tube, 2mL of chloroform was added, while the 3mL of conc. H_2SO_4 was equally added to form a layer. Formation of reddish-brown colour indicates the presence of terpenoids [1].

2.8.5 Test for Tannins

The presence of tannin was determined by adding a few drops of 0.1% ferric chloride to 1 mL of solvent extract. Brownish green or blue black coloration was formed [10].

2.9 Assessment of Disease Severity (ADS)

Symptoms of infection such as disappearance of leaf veins, yellowing and stunting of older leaves, drooping of petioles were observed and recorded at 8th week of planting. The measurement of disease severity was based on a scale of 1-6 as shown in Table 1.

Table 1: Disease severity scale for rating *Capsicum chinense* plant

Scale	Interpretation
0	Healthy plant, no symptoms
1	Symptoms at leaflets tips
2	Very few streaks on old leaves gradually decreasing on young leaves
3	Moderate streaking on leaflets
4	Half of the plant wilted
5	More than half leaf area (60-70%) wilted
6	Severe wilting on more than 70% leaf area Plants wilted or dead

2.10 Statistical Analysis

The data collected was subjected to analysis of variance (ANOVA) in order to find out if there were significant differences among the treatments, the least significant difference (LSD) was calculated due to the possible level of significance that may be observed by using SPSS.

3.0 Results

3.1 Analysis of Soil

The physicochemical properties of soil Used for the greenhouse experiment are shown in Table 2. The soil was observed to have 13% clay, 82.5% sand and 4.5% silt. The available sodium, calcium, magnesium, and potassium contents are 1.14, 2.59, 3.13, 3.14 cmol/kg respectively while the amount of Iron, manganese, zinc, and phosphorus are 13, 24.8, 6.7 and 13.11 mg/kg respectively. The soil had optimal pH of 7.21 for plant growth and showed low organic carbon (1.26), organic matter (2.17) as well as low total nitrogen (0.11%) as presented in Table 2.

Table 2: Soil Analysis

Sample parameters test	Values
pH (H_2O)	7.21
O.C	1.26
O.M	2.17
TN %	0.11
Na (Cmol/kg)	1.41
Ca (Cmol/kg)	2.59
Mg (Cmol/kg)	3.13
K (Cmol/kg)	3.13
Fe (mg/kg)	13
Cu (mg/kg)	0.6
Mn (mg/kg)	24.8
Zn (mg/kg)	6.7
P (mg/kg)	13.11
% Sand	82.5
% Clay	13
% Silt	4.5

3.2. Phytochemical Screening of the leaves of *C. chinense*

The phytochemical analysis from this study revealed the presence of saponin, flavonoid, tannin and alkaloid in the leaves of all the treatments of *Capsicum chinense*. It was also observed that the control contains all tested physiological parameters but lacks terpenoid, which was absent in all examined plants (Table 3).

Table 3: Phytochemical constituents of *Capsicum chinense* leaves

Phytochemical compd Treatments	Saponins	Flavonoids	Alkaloids	Terpenoids	Tannins
Control	+	+	+	-	+
P1C1	+	+	+	-	+
P1C2	+	+	+	-	+
P2C1	+	+	+	-	+
P2C2	+	+	+	-	+
T2C1	+	+	+	-	+
T2C2	+	+	+	-	+
P1C1 + T2C1	+	+	+	-	+
P1C1 + T2C2	+	+	+	-	+
P1C2 + T2C1	+	+	+	-	+
P1C2 + T2C2	+	+	+	-	+
P2C1 + T2C1	+	+	+	-	+
P2C1 + T2C2	+	+	+	-	+
P2C2 + T2C1	+	+	+	-	+
P2C2 + T2C2	+	+	+	-	+

Keys: + = Present, - = Absent

3.3. Assessment of disease severity

The result obtained as presented in Table 4 shows that the disease severity of the plants treated with only the pathogens were significantly higher than the control. Infected plants treated with low concentration of the antagonist (combined treatment) had slight symptoms. However, there was no symptoms of disease observed as the concentration of the antagonist's increases. Likewise, plants treated with the antagonist alone were very healthy with no symptoms.

The diseased severity showed scales of 3,4,3,5,0,0 for P1C1,P1C2,P2C1,P2C2,T2C1 and T2C2 respectively, while 1,0,1,1,1,0,1,1,0 for P1C1+T2C1, P1C1+T2C2,P1C2 +T2C1,P1C2+T2C2 +P2C1 + T2C1,P2C1 +T2C1 and P2C1 +T2C2 respectively as presented in Table 4.

Table 4: Assessment of biocontrol effects of *Trichoderma koningii* on the disease severity on a scale of 1–7 for *Fusarium oxysporum* and *Pythium ultimum* on *Capsicum chinense*.

Treatments	Severity
Control	1
P1C1	3
P1C2	4
P2C1	3
P2C2	5
T2C1	0
T2C2	0
P1C1 + T2C1	1
P1C1 + T2C2	0
P1C2 + T2C1	1
P1C2 + T2C2	1
P2C1 + T2C1	1
P2C1 + T2C2	0
P2C2 + T2C1	1
P2C2 + T2C2	0

3.4Antagonistic effect of *Trichoderma koningii* against *Fusarium oxysporum* and *Pythium ultimum* as observed in *Capsicum chinense* In-vivo

The bioagent *Trichoderma koningii* was observed to have significant antagonistic effect against *Fusarium oxysporum* and *Pythium ultimum* inoculated on *Capsicum chinense* by studying the agronomic parameters of the plant as presented in Figures2 -8. It was also observed that the treatments (bioagent and pathogens) had different effect at varying levels of concentration.

From Figure2, it was observed that treatments with *Trichoderma koningii* at concentration 1.0×10^7 (T2C2) had the highest plant height (25.00 cm) followed by P1C2T2C2 (24.00cm) and T2C1 (22.5cm) compared to the control 17.5 cm. Also plants treated with both the pathogens and antagonist (P1T2) had higher plant height as compared to plants treated with pathogens alone at different concentrations. However, the lowest plant height was observed in treatments with the pathogens.

The result presented in Figure 3 showed that Plants treated with the bioagent significantly increases the number of leaves of *Capsicum chinense*. It was observed that T2C2, T2C1 and P1C1T2C2 had higher number of leaves than the control. Likewise, all treatments with *Trichoderma koningii* had higher number of leaves as against plants treated with pathogen alone which had the lowest number of leaves.

It was also observed in Figure 4 that the leaf area of plants treated with *Trichoderma* was higher than the untreated plants while the lowest leaf area was recorded in plants infected with the pathogen P1C2 (2.00cm).

The stem girth of all *Trichoderma* treated plants are not significantly different from each other at $P \leq 0.05$. It was observed that stem girth of P1C1T2C1, P1C1T2C2, P1C2T2C1, P1C2T2C2, P2C1T2C1, P2C1T2C2, P2C2T2C1 and P2C2T2C2 are 0.26, 0.26, 0.25, 0.28, 0.30, 0.30, 0.27, and 0.32cm as recorded in Figure 5. However, these are significantly higher than the stem girth of plants treated with the pathogens only. Petiole length measurements in Figure 6 revealed that treatments with *Trichoderma koningii* significantly affected the petiole length of *Capsicum* plant. T2C2 and T2C2 had highest petiole length of 2.30 and 2.35cm respectively followed by the untreated plants (control) with petiole length of 1.8 cm. The combined treatments of pathogen and *Trichoderma* (e.g. P1C2T2C2) shows higher petiole length when compared to plants treated with pathogen.

In Figure7, the highest number of leaves was recorded in P1C2T2C2 and control 5.00cm followed by T2C1, P1C1T2C1, P1C1T2C2, P1C2T2C1, P2C1T2C1, P2C1T2C2, P2C2T2C1 and P2C2T2C2 which have the same number of leaves 4.00cm. The highest internode length was recorded in T2C2 (5.50cm) while the lowest was recorded in P2C2 (1.90cm) as compared to the control (3.00cm).

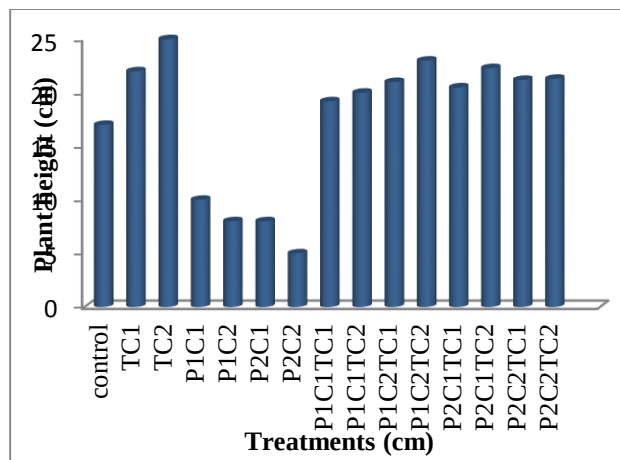


Figure 1: Effect of *Trichoderma koningii* on *Fusarium oxysporum* and *Pythium ultimum* as shown in plant height of *Capsicum chinense*.

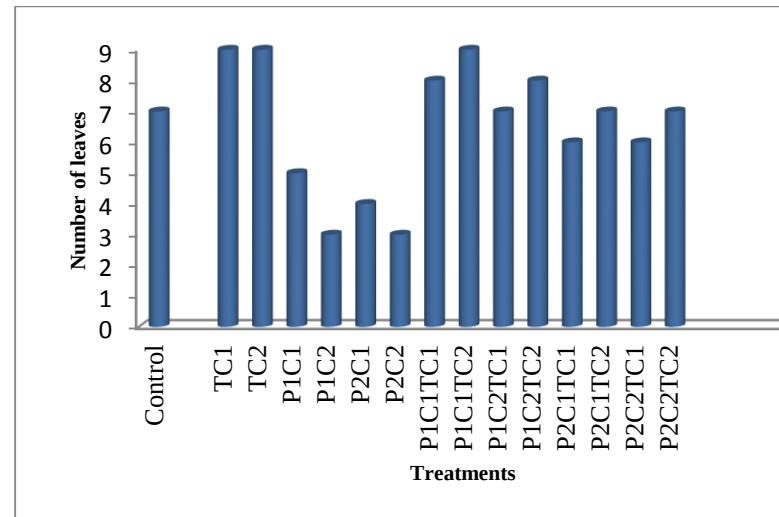


Figure 2: Effect of *Trichoderma koningii* against *Fusarium oxysporum* and *Pythium ultimum* as observed in the number of leaves of *Capsicum chinenses*.

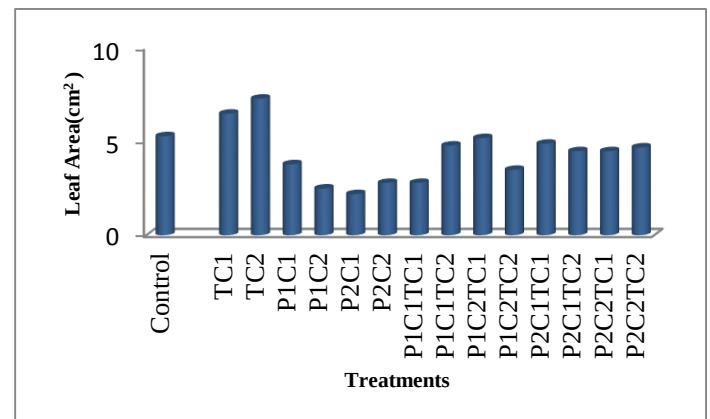


Figure 3: Effect of *Trichoderma koningii* on *Fusarium oxysporum* and *Pythium ultimum* as recorded on leaf area of *Capsicum chinense*

and *Pythium ultimum* as observed in the number of branches of *Capsicum chinense* plant .

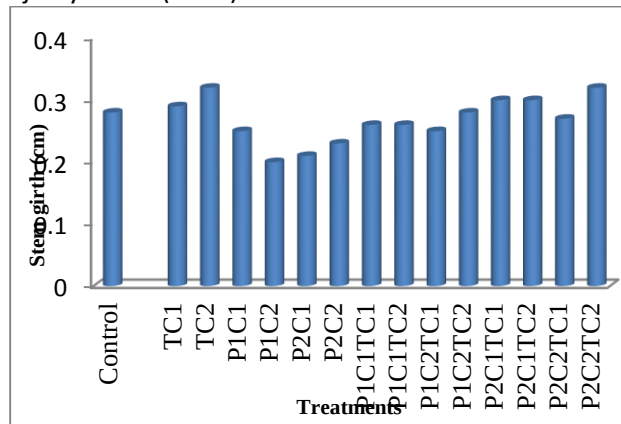


Figure 4: Effect of *Trichoderma koningii* on *Fusarium oxysporum* and *Pythium ultimum* as observed in stem girth of *Capsicum chinense*

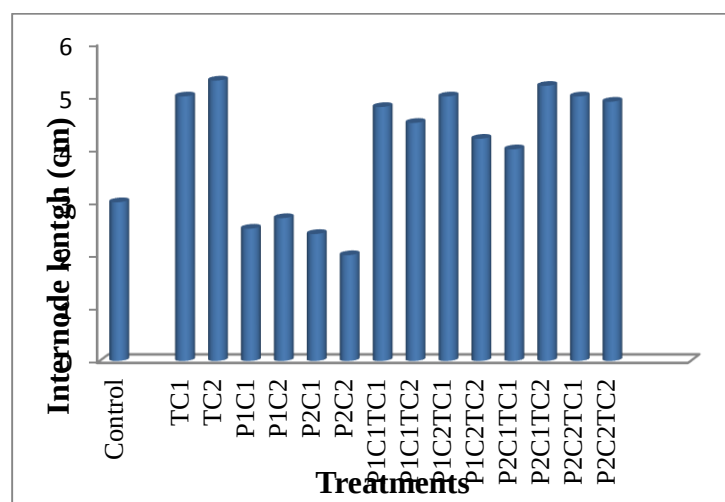


Figure 7: Effect of *Trichoderma koningii* on *Fusarium oxysporum* and *Pythium ultimum* internode length of *Capsicum chinense* plant

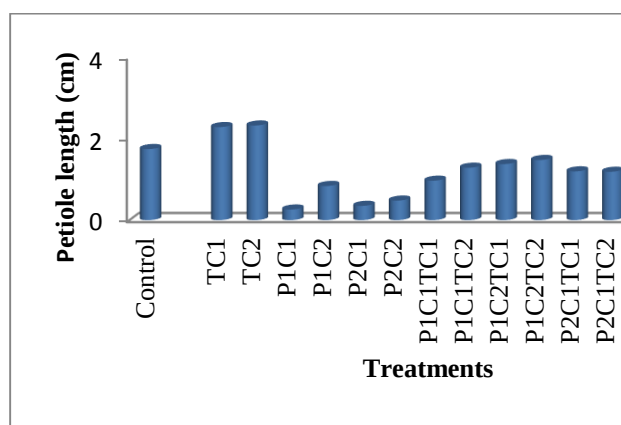


Figure 5: Effect of *Trichoderma koningii* on *Fusarium oxysporum* and *Pythium ultimum* as shown in petiole length of *Capsicum chinense* plant

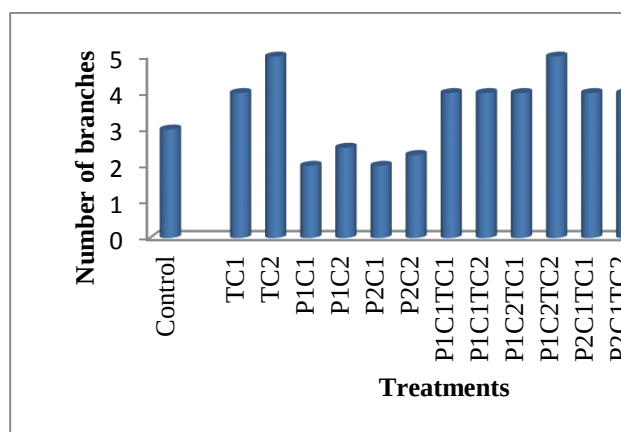


Figure 6: Effect of *Trichoderma koningii* on *Fusarium oxysporum*

Where,
TC1 = *T. koningii* (1.0×10^5) spore/mL, TC2 = *T. koningii* (1.0×10^7) spore/mL,

P1C1 = *F.oxysporum* (1.0×10^5) spore/mL, P1C2 = *F. oxysporum*(1.0×10^7) spore/mL,
P2C1 = *P. ultimum* (1.0×10^5) spore/mL, P2C2= *P.ultimum*(1.0×10^7) spore/mL,
P1C1TC1 = *F. oxysporum* (1.0×10^5) spore/mL + *T.koningii* (1.0×10^5) spore/mL,
P1C1TC2 = *F. oxysporum* (1.0×10^5) spore/mL + *T. koningi* (1.0×10^7) spore/mL,
P1C2TC1 = *F. oxysporum* (1.0×10^7) spore/mL + *T.koningii* (1.0×10^5) spore/mL,
P1C2TC2 = *F. oxysporum* (1.0×10^7) spore/mL + *T.koningii* (1.0×10^7) spore/mL,
P2C1TC1 = *P. ultimum* (1.0×10^5) spore/mL + *T.koningii* (1.0×10^5) spore/mL,
P2C1TC2 = *P. ultimum* (1.0×10^5) spore/mL + *T.koningii* (1.0×10^7) spore/mL,
P2C2TC1 = *P. ultimum* (1.0×10^7) spore/mL + *T.koningii* (1.0×10^5) spore/mL, and
P2C2TC2 = *P. ultimum* (1.0×10^7) spore/mL + *T.koningii* (1.0×10^7) spore/mL,

4.0 Discussion

Trichoderma species are commonly utilized as biocontrol agents because they have multiple benefits on plant, including stimulating plant growth, boosting nutrient intake from the soil, and decreasing the activity of soil borne diseases, all of which have an impact on plant development[9],[17]. Trichoderma spp. have evolved numerous mechanisms like mycoparasitism, production of inhibitory substances, inactivation of pathogen enzymes and induction of resistance to attack other fungi and enhance plant and root growth [11].

The result obtained from the phytochemical analysis showed the presence of some phytochemical components in *Capsicum chinense* such as tannin, saponin, alkaloids, terpenoids and flavonoids. This implies that phytochemical compounds contained in the leaves of *Capsicum chinense* plant were not altered by the different treatments applied[18], [19], [10], [3]. Phytochemicals such as tannins, saponins, flavonoids, phenols, alkaloids and volatile oils are present except terpenoids. Phytochemical polyphenols have received great attention because of their biological activity. Many flavonoids have shown strong antioxidant properties [20]. These results indicates that the pepper containing high phenolics provides a source of antioxidants and in addition it improves flavour to food hence it is used as a value added ingredient for stabilizing food against lipid peroxidation reactions[11]

This study gives a strong evidence of the potential ability of Trichoderma species to enhance plant height (cm), shoot length(cm), number of leaves, nodal length(cm), petiole length (cm), number of branches, leaf area(cm²) and stem girth(cm)[17] and also to suppress the growth of *F. oxysporum* and *P. ultimum* inoculated into *C. chinense* [21]. The study on plant height of *Capsicum chinense* shows that *T. koningii* significantly ($p < 0.05$) promote the growth of the plant as observe in all treatments with the bioagent. Investigations carried out by Lo and Lin [22] on plant and root growth of cucumber and bitter gourd reveals that Trichoderma strains significantly increased seedling height by 26 to 61%.

Trichoderma koningii had also been found to have high significant impact on the number of leaves of rice plants compared to pathogen treatment and

control [15], this conform with the recent research findings in *C. chinense*. And it was observed that Inoculation with *T. Koningii* at C2 (1.0×10^7) had the highest leaf number. The possession of more number of leaves increases the rate at which photosynthesis occurred in plant leading to enhanced growth performance, therefore plant treated with *Trichoderma koningii* undergo photosynthesis more when compared to plant that have lesser leaf numbers recorded in the ones treated with *Pythium ultimum* and *Fusarium oxysporum*.

The leaf area of *C. chinense* was considerably enhanced with *T. Koningii* as compared to treatment with the pathogens. Leaf area determines light interception, energy conversion and water balance. It is a reliable parameter for determining plant productivity [22].

The higher significant effect of some of the physiological parameters (e.g. petiole length and number of branches) obtained in the results conform with the findings of [23]. It was discovered that the higher significant effect found in the leaf number and petiole length of *Dialium guineensis* and *P. africana* seedlings might be due to higher phosphorus content in the soil. According to [24] phosphorus promotes healthy root growth and biomass accumulation. Higher content of nitrogen further enhances plant height, number of leaves, stem girth, number of branches, internode length, leaf area and number of leaf branches. Another scientists [25] further investigated the role and responsiveness or relative field mycorrhizal dependency (RFMD) as the extent to which a plant requires mycorrhizal inoculation to produce maximum growth rate for the seedlings.

The inoculation of Trichoderma organisms against the pathogenic organism affecting *C. chinense* enhanced the number of branches. This gave higher yield by application of Trichoderma species in mustard, wheat, corn, tuberose, sugarcane, tomato, okra[26], [16], [11], [14], it was also found to be the same with *Capsicum chinense*.

However, the growth rate of *C. chinense* were influenced by disease severity caused by the pathogenic organisms[4]. The research conducted in Kenya on disease incidence, severity and yield loss of finger millet varieties and invitro mycelia growth inhibition of *P. grisea* using biological antagonists and fungicides, revealed that finger millet blast were higher at the station when compared to farmers' field

in Busia and Tesodistricts with incidence range of 2-9(mean 3.5) and severity range of 10-90%. The survey revealed that the disease incidence and severity varied from locality to locality due to cultivation of different varieties in different geographical and environmental conditions prevailing in each locality [27].

In agreement with the current research findings, the highest disease severity (5) was discovered in concentration 2 of *P. ultimum* (P2C2). This may be associated with the level of damage caused by the organism not only to the shoot but also to the root of *C. chinense*. The affected part of the plant may reduce the absorption of light intensity and photosynthetic capability. Increase in relative humidity favours the spread and the distribution of the wilting pathogen, *F. oxysporum* which had high disease severity of 4 at concentration 2 (P1C2). However, low relative humidity that may not favour the spread and the distributions of the disease which in turn leads to a better growth yield of the plant under investigation [19]. It was observed in the study that environmental and technical factors that are not related to fungal or bacterial organisms are also determining factors that can influence the growth of the plant [21].

5.0 Conclusion

Trichoderma koningii is a good biocontrol agent in controlling the *Pythium ultimum* and *Fusarium oxysporum* in *Capsicum chinense*, alongside with the diseases caused by the pathogens. It also improved the growth of pepper plant, this was obvious in some of the physiological parameters that was investigated. Plant treated with *Trichoderma koningii* had higher percentage of growth at higher concentration compared to those treated with pathogens. Therefore, *Trichoderma koningii* is a good biocontrol agent that can work against *Fusarium oxysporum* and *Pythium ultimum* in place of chemical fungicides which are hazardous and expensive.

Declarations

Ethics approval and consent to participate

Not applicable

Consent for publication

Not applicable

Availability of data and materials

All datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

M.D.A developed the concept of the research, collected, analysed the data and wrote the manuscript., A.A.S provided guidance throughout the study and corrected the manuscript in line with the journal guideline.

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Not applicable

6.0 References

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